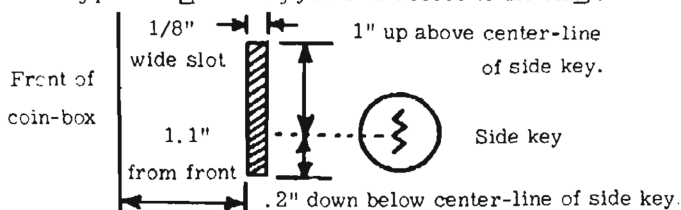
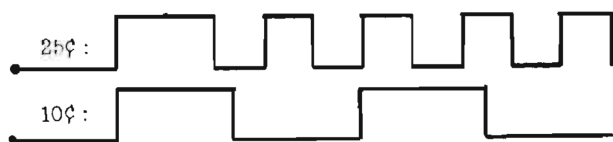


3. The following cutting pattern would be better than a drilling pattern providing you have access to the side:



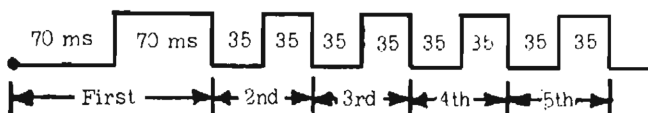
4. The tones' intervals are not generated by the coins pushing levers on the red box mechanism (like everyone thinks) but by a little relay which reverses the direction of the wheel that the levers cock. In other words, when a coin is dropped, the coin pushes a lever which cocks a wheel. The coin then drops into the drop chute, and the red box goes thru a 70ms delay to allow the coin to clear the mechanism. After that, a unijunction-timer circuit pulls in a relay which unwinds the wheel until it is back where it started. The number of times the relay ticks, pulling the wheel, is the number of beeps. If the wheel went 5 notches forward, a switch drops, telling the unit to do 35ms beeps (for a quarter). However, the first beep will always be the same as the above-mentioned 70ms pre-beep delay, because the coin will just be clear, and all of the pulses (beeps) are actually symmetrical square waves:



1. Coin Drop Mechanism - In my last letter, I asked about the "coin-sense relay" but couldn't figure how Bell knew about the money. Now I know! First, they use a low current (20ma) high voltage (50v) relay to release the money. Second, the relay resets a vane which senses the presence of the money. This vane cannot be checked by the operator. However, she can see the current drain caused by the drop relay pulling in. If the vane hasn't been hit by money, the coin-drop relay won't pull in even when the operator tries to make it. The good part is that she can't check the vane until she drops the money. So it doesn't matter when you drop your nickel, as long as you drop one before each time she collects. Also, she can't check it on

NO. 49

3. As I told you last time, the 25¢ set of tones are 5 beeps, the first one twice as long as the rest:



Mugs

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THE PRINCIPLES OF TWX PHREAKING

by
Cheshire

INTRODUCTION

IN 1971, THE WESTERN UNION TELEGRAPH CO. BOUGHT THE TELETYPE/TELEX EXCHANGE & TELEGRAPH CO. UNDER THIS AGREEMENT ATXT COULD NOT ENTER DATA COMMUNICATIONS UNDER 100 BAUD (10 CHARACTERS PER SECOND) UNTIL APRIL 1986. MEANWHILE, WHY WOULD ANY OF THE INDEPENDENT TELECO. BOTS WHILE CHW EXCHANGES FOR THIS PULL PLUNKY? THEY WOULDN'T. OF COURSE, THEY TRANSLATE FROM FUNNY TWX AREA CODES TO NORMAL TELEPHONE EXCHANGES. FIND THE RIGHT EXCHANGE, AND NIN THE TWX NETWORK.

TRANSLATING TWX EXCHANGES

THERE IS AN EASY WAY OF TRANSLATING A TWX EXCHANGE. THAT'S WHEN IT IS "GIVEN" TO YOU IN THE TWX DIRECTORY. TWX AREA CODES ARE USUALLY 910, 910, 710, 800, & 910 IN THE UNITED STATES, AND 410 IN CANADA. SOME TWX MACHINES HAVE REGULAR TELEPHONE NUMBERS, BECAUSE THEY WERE INSTALLATIONS BEFORE THE MODEL 32 TELETYPE BECAME THE STANDARD OF THE TWX NETWORK. THESE OLDER MACHINES HANDLE THEIR OWN LIMITED SPEED CONVERSION, AND COMMUNICATIONS IS POSSIBLE BETWEEN THESE OLD MACHINES, AND THE NEWER MODELS.

WORKING WITH A MODEL 32 (OR EQUIVALENT) COMPUTER TERMINAL SUCH AS A COMPUTER DEVICES 1103, IT IS SIMPLE. (A) OF TELETYPE/TELEX HT-20 CAN SEND TO A TWX TERMINAL. A TWX MACHINE ANSWERS WITH THE SAME "CARRIER" TONE AS A COMPUTER DOES. SO YOUR TERMINAL WILL LIGHT ITS CARRIER LIGHT (IF IT HAS ONE). EACH MODEL 32 HAS A PROGRAMMED "ANSWERBACK" WHEN, IN IT, IT IS PROGRAMMED WITH AN IDENTIFIER OF 30 CHARACTERS OR LESS. THE FIRST THREE CHARACTERS ARE USUALLY VARIANCE RETURN (CR), LINE FEED (LF), AND PAUSE (PA). THE LAST THREE ARE USUALLY, CR, LF, CONTROL D. THE CONTROL A CHARACTER WILL TURN ON THE PAPER TAPE READER OF THE DISTANT TERMINAL (IF IT IS PROGRAMMED TO DO SO), WHICH MOST ARE. THIS LEAVES 14 CHARACTERS LEFT FOR IDENTIFICATION. AN EXAMPLE WOULD BE: MIDLINGPFF DET. THIS IS THE ANSWERBACK OF THE DETROIT WINGS OF THE NATIONAL HOCKEY LEAGUE. AND WOULD FURTHER

SEEM TO BE THEIR PR DEPARTMENT

THE TWX NUMBER OF THE WINGS IS 910-221-5032 AS WE LOOK DOWN THE LISTINGS FOR DETROIT IN A 1974 OR EARLIER TWX DIRECTORY. LATER ONES LUMP A WHOLE STATE TOGETHER. WE FIND LISTINGS FOR OTHER INSTALLATIONS. THE EXCHANGES FOR THESE OLDER MACHINES ARE 311-538, 312-222, 313-342, AND 313-777. WHEN WE HIT 222-5032 WE FIND THE ANSWERING TIME OF A TWX MACHINE. OTHER TWX MACHINES IN DETROIT ARE 910-221-3127 (DETROIT BO DET), 910-221-5030 (DET), AND 910-221-5171 (DET FANN DET).

SET IS, OF COURSE, THE CITY CODE OF DETROIT. OTHER CITY CODES ARE 910 FOR NEW YORK, 410 FOR LOS ANGELES, 910 FOR SAN FRANCISCO, AND 600 FOR CHICAGO. SMALL COMMUNITIES HAVE FOUR LETTER CITY CODES. I DON'T KNOW THE TH: TRANSLATIONS FOR ALL THE ABOVE CITIES, BUT THOSE SENDING TWX TRANSLATIONS TO TAP WILL GET A LIST OF THE FEW CITIES TRANSLATIONS ARE AVAILABLE FOR. I HAVE TRANSLATED CITIES WITH TAPCO, STOPS, AND SENT MYSELF MESSAGES WHEN TRAVELING.

ANOTHER POSSIBILITY IS TO MAKE FRIENDS WITH SOMEONE WHO WORKS IN A SMALL OFFICE FOR A BRANCH OFFICE OF A LARGER COMPANY, AND USE THEIR TWX MACHINE. THE TRICK THEN IS TO CALL A TWX NUMBER THAT DOES NOT EXIST AND LET THE PHONE COMPANY COMPUTER TELL YOU WHAT NUMBER IT THINKS YOU WANTED. THIS WORKS FOR VERY FEW CITIES.

EXAMPLE: THE WESTERN UNION TELEGRAPH CO. IN UPPER MERLE RIVER, NJ IS 710-998-2251. IF WE DIAL 910-998-2251 ON A TWX MACHINE, WE'LL HEAR A REPEATING BAY. "THE NUMBER YOU HAVE DIALLED 779-2251" WHICH TELLS OF THAT BY DIALING 710-998-2251 WE CAN REACH NJ 998-2251. IF WE THEN LOOK CLOSELY AT THE NUMBER GIVEN FOR THE FIVE NEWS SERVICE REFERRED TO TWO SUBSCRIBERS, THE NUMBER GIVEN IS 910-998-9999. BY DIALING 910-998-9999 AND THROWING THE HAND-SET INTO THE ACOUSTIC COUPLER, OUR TERMINAL THINGS OUT "HUI FUI HUI" (THE ANSWERBACK OF THE SYSTEM). A COMPUTER LOCATED IN MANHATTAN, THIS MACHINE IS UPDATED ONCE AN HOUR BY UP OVER DEDICATED RADIO-UP LEASED LINES. AS THE COMPUTER SENDS ITS ANSWERBACK, IT TRANSMITS CONTROL E. CONTROL E SHOULD TRIP THE ANSWERBACK OF THE CALLING TWX MACHINE SINCE THE COMPUTER WILL GO NO.

FURTHER UNTIL IT RECEIVES SOMETHING. TYPE IN ANY RANDOM CHARACTERS (WHICH IT WILL LET YOU IN TYPE CATALOG) OR JUST "CATY" AS IT ONLY SPEAKS THE FIRST FOUR CHARACTERS AFTER A SPACE. YOU CAN TYPE "HUI FUI HUI STOP CR LF" AND IT WILL GIVE YOU THE LATEST NEWS. WEATHER FOR 20 CITIES, AND SPORTS. THE CATALOG GIVES INSTRUCTIONS AND TELLS YOU WHAT INFORMATION IS AVAILABLE. THIS IS A "DIAL-A-R-ONE" TYPE SERVICE. PAID FOR BY THE LINE CHARGES OF THE LONG DISTANCE CALL IN REACHING IT.

IT SHOULD BE NOTED THAT YOUR TERMINAL SHOULD BE SET FOR 110 BAUD, AND EVEN OR NO PARITY. HAVE FUN, HAPPY HACKING, AND REMEMBER

THE #4 PHREAKER

ATTN: MILO FONEBILL

Please get in touch with me through the TAP office.

Tom Edison

COMING NEXT ISSUE !!!

Our next issue, #80, marks the 7th anniversary of TAP. To mark the occasion, issue #80 will be a special double edition with color pages. I want this to be the best fuckin' issue of TAP that we've ever put out! I'm asking all of our readers, especially those who have sat in their latrasses and done nothing to help out over the past years, to send in information, newsclips, articles, and columns. I want 4 pages just bursting with technical knowledge. Let's show Bell that we DO care!

LETTERS FROM READERS

Dear TAP,

Perhaps some of your readers would like to chat with ships on the high seas. This is now possible through a satellite called Marisat. Actually there are two currently operational: Atlantic and Pacific, and in a very short time there will be one for the Indian Ocean too. Enclosed is a list of most of the ships that you can contact and the list continues to grow. To place a call, engage your favorite 800 or 855 number, then box off to 800-243-3640 (or 284-9393 in Conn.). This gets you the Marisat operator. She will ask you billing information, so you should have a story made up as to who (what individual/or company) is calling and give her a number to bill it to (like IBM or Continental Bakery, or whatever). She doesn't have any way (presently) of knowing where the call is coming from. (It's going to cost somebody 10 bucks a minute!) And be a good guy and give her the ship's number (like SS Jackson #1883201) and tell her you want to speak with..... oo, what the hell, the chief steward or cabin boy - anyway, the radio operator will get the call and you can bullshit with him. Sparks can tell you where the ship is, courses, speed, destination, etc.

Comsat has some other numbers that might prove fruitful if somebody knows how to exploit them: (202) 264-8333 is Data Test (associated digits 1800-822-4646, which may be a computer entry code); 264-8340, 264-8330 (data sen), and 264-8330. You might also try 284-4391/4382.

Has anybody ever "boxed" calls through TWX? I don't know if it's a usable system. (See article on TWX phreaking by Osbert Kallgren in this issue of TAP.) Anyway Marisat has Telex capability too: 710-381-8620 (toll free). Additionally, AT&T long lines is 914-382-8282 and the United Nations language bank (free translations?) is 312-985-3361.

May the force be with you,
R2-D2

Greetings,

In Missouri, Southwestern Bell has forced itself upon an unwilling public. They now charge 20¢ per directory assistance call (after the first five free ones per month).

Well, in a burst of insight it came to me--and it has been verified by experiment and questions to operators. It is:

Call the 555-1212 number, using the 1 code, omitting the area code (as in calling directory assistance for another town in your area code). When the operator asks you "What city please?" just say the name of your town (in my case St. Louis)--and she says "go ahead." They have no way of knowing what town you are dialing from! And there is no charge for the 555-1212 number.

I also, note that Kansas City, Mo 555-1212 info uses computers to retrieve telephone numbers. When will I be able to access by computer the up to date directory listings?

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US First Class in plain sealed envelope \$7.

Canada & Mexico First Class \$7.

Foreign \$8.

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Back - Monthly \$1.00.

Electronic charges - 10¢ each. A - DC Basics.

B - AC Basics, C - Phone Basics, D - Amplifiers.

TAP mugs - \$4.00.

TAP T-shirts - \$4.00. Specify size and color: Small, Medium, Large, Extra Large, Black or red T-shirt with white TAP logo.

TAP TAP T-shirts - \$4.00. Specify size and color: Medium, Large, Extra Large, Blue, tan, or grey.

Send all orders to my sister (M. Cash) at:

TAP, P.O. Box 410, 40 West 14 Street, New York, N.Y. 10011

and a copy to:

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TAP, P.O. Box 410, 40 West 14 Street, New York, N.Y. 10011

Shipboard Terminal Installations: An Update

At this time, the following vessels were equipped with shipboard terminals which have been confirmed for operation with the Marisat System.

Ship	Country of Registration	Ship Number	Antennas
Alexis	France	130171	4000 130171
Alexis Star	United States	150177	4000 150177
American Ace	United States	150178	4000 150178
Aquarius	United States	150179	4000 150179
Arc Adventure	United States	150180	4000 150180
Arc Encounter	United States	150181	4000 150181
Arc Venture	United States	150182	4000 150182
Arc Venture Bay	United States	150183	4000 150183
Arcadia	Denmark	201010	4000 201010
Ben Ocean Liners	United Kingdom	204000	4000 204000
Cable Venture	United Kingdom	204001	4000 204001
Chalenger	United States	150184	4000 150184
Coral	United States	150185	4000 150185
Coral Reef	United States	150186	4000 150186
Coral Reef	United States	150187	4000 150187
Coral Reef	United States	150188	4000 150188
Coral Reef	United States	150189	4000 150189
Coral Reef	United States	150190	4000 150190
Coral Reef	United States	150191	4000 150191
Coral Reef	United States	150192	4000 150192
Coral Reef	United States	150193	4000 150193
Coral Reef	United States	150194	4000 150194
Coral Reef	United States	150195	4000 150195
Coral Reef	United States	150196	4000 150196
Coral Reef	United States	150197	4000 150197
Coral Reef	United States	150198	4000 150198
Coral Reef	United States	150199	4000 150199
Coral Reef	United States	150200	4000 150200
Coral Reef	United States	150201	4000 150201
Coral Reef	United States	150202	4000 150202
Coral Reef	United States	150203	4000 150203
Coral Reef	United States	150204	4000 150204
Coral Reef	United States	150205	4000 150205
Coral Reef	United States	150206	4000 150206
Coral Reef	United States	150207	4000 150207
Coral Reef	United States	150208	4000 150208
Coral Reef	United States	150209	4000 150209
Coral Reef	United States	150210	4000 150210
Coral Reef	United States	150211	4000 150211
Coral Reef	United States	150212	4000 150212
Coral Reef	United States	150213	4000 150213
Coral Reef	United States	150214	4000 150214
Coral Reef	United States	150215	4000 150215
Coral Reef	United States	150216	4000 150216
Coral Reef	United States	150217	4000 150217
Coral Reef	United States	150218	4000 150218
Coral Reef	United States	150219	4000 150219
Coral Reef	United States	150220	4000 150220
Coral Reef	United States	150221	4000 150221
Coral Reef	United States	150222	4000 150222
Coral Reef	United States	150223	4000 150223
Coral Reef	United States	150224	4000 150224
Coral Reef	United States	150225	4000 150225
Coral Reef	United States	150226	4000 150226
Coral Reef	United States	150227	4000 150227
Coral Reef	United States	150228	4000 150228
Coral Reef	United States	150229	4000 150229
Coral Reef	United States	150230	4000 150230
Coral Reef	United States	150231	4000 150231
Coral Reef	United States	150232	4000 150232
Coral Reef	United States	150233	4000 150233
Coral Reef	United States	150234	4000 150234
Coral Reef	United States	150235	4000 150235
Coral Reef	United States	150236	4000 150236
Coral Reef	United States	150237	4000 150237
Coral Reef	United States	150238	4000 150238
Coral Reef	United States	150239	4000 150239
Coral Reef	United States	150240	4000 150240
Coral Reef	United States	150241	4000 150241
Coral Reef	United States	150242	4000 150242
Coral Reef	United States	150243	4000 150243
Coral Reef	United States	150244	4000 150244
Coral Reef	United States	150245	4000 150245
Coral Reef	United States	150246	4000 150246
Coral Reef	United States	150247	4000 150247
Coral Reef	United States	150248	4000 150248
Coral Reef	United States	150249	4000 150249
Coral Reef	United States	150250	4000 150250
Coral Reef	United States	150251	4000 150251
Coral Reef	United States	150252	4000 150252
Coral Reef	United States	150253	4000 150253
Coral Reef	United States	150254	4000 150254
Coral Reef	United States	150255	4000 150255
Coral Reef	United States	150256	4000 150256
Coral Reef	United States	150257	4000 150257
Coral Reef	United States	150258	4000 150258
Coral Reef	United States	150259	4000 150259
Coral Reef	United States	150260	4000 150260
Coral Reef	United States	150261	4000 150261
Coral Reef	United States	150262	4000 150262
Coral Reef	United States	150263	4000 150263
Coral Reef	United States	150264	4000 150264
Coral Reef	United States	150265	4000 150265
Coral Reef	United States	150266	4000 150266
Coral Reef	United States	150267	4000 150267
Coral Reef	United States	150268	4000 150268
Coral Reef	United States	150269	4000 150269
Coral Reef	United States	150270	4000 150270
Coral Reef	United States	150271	4000 150271
Coral Reef	United States	150272	4000 150272
Coral Reef	United States	150273	4000 150273
Coral Reef	United States	150274	4000 150274
Coral Reef	United States	150275	4000 150275
Coral Reef	United States	150276	4000 150276
Coral Reef	United States	150277	4000 150277
Coral Reef	United States	150278	4000 150278
Coral Reef	United States	150279	4000 150279
Coral Reef	United States	150280	4000 150280
Coral Reef	United States	150281	4000 150281
Coral Reef	United States	150282	4000 150282
Coral Reef	United States	150283	4000 150283
Coral Reef	United States	150284	4000 150284
Coral Reef	United States	150285	4000 150285
Coral Reef	United States	150286	4000 150286
Coral Reef	United States	150287	4000 150287
Coral Reef	United States	150288	4000 150288
Coral Reef	United States	150289	4000 150289
Coral Reef	United States	150290	4000 150290
Coral Reef	United States	150291	4000 150291
Coral Reef	United States	150292	4000 150292
Coral Reef	United States	150293	4000 150293
Coral Reef	United States	150294	4000 150294
Coral Reef	United States	150295	4000 150295
Coral Reef	United States	150296	4000 150296
Coral Reef	United States	150297	4000 150297
Coral Reef	United States	150298	4000 150298
Coral Reef	United States	150299	4000 150299
Coral Reef	United States	150300	4000 150300

Dear Joe:

I understand that at some point in the recent past your readers, or your magazine itself were interested in knowing the frequencies for those digits that come after 9.

They are: 0 (10) 941.0 + 1356.0
 * (11) 941.0 + 1209.0
 * (12) 941.0 + 1477.0
 * (13) 697.0 + 1633.0
 * (14) 770.0 + 1633.0
 * (15) 852.0 + 1633.0
 * (16) 941.0 + 1633.0
 * (17) 350.0 + 440.0

this information is available through many publications e.g. Bracco Controls Division, Leduc Inc., College and South streets, Lima Ohio 45356, (513) 773-6271. I trust this information will not be used for anything illegal or foolhardy. Bracco's booklet could give you more information.

one lonely Canadian



Phone Call Is Rated X

Seattle (UPI) — It may work for Superman, but changing your clothes in a public telephone booth can mean trouble for the rest of us.

Huber Martin, 31, stepped into a glass phone booth next to a public park and exposed himself while changing trousers, police said.

He was arrested for changing into blue denim cutoffs while 25 to 30 persons of both sexes looked on.

FLASH! LATE REPORT!

KON-KLAVE KANCELLED

by Cheshire

There was to be a Phone Phreak Konklave at the New York Coliseum running concurrently with the Personal Computing Expo (PCX) over the Halloween weekend. About 1/4 of the TAP mailing list (those in the "Bus-Wash" and a few west-coasters) got the word a week or so before the event. The evening before the event was to begin, I was approached by people from Byte magazine, a co-sponsor of PCX. It should be explained that all negotiations had been handled by H A Bruno Associates in New York, who had arranged to rent the Coliseum for the show. I had already checked that no nearby hotel had function rooms (I didn't bother checking the Central Park South hotels), and approached Bruno Assoc. as a representative of a Communications Hobbyist Group wishing to rent function space. When the folks from Byte came into town, they had a fit.

In June 1975, 73 magazine published the schematic diagram for a Blue Box. Pa Bell came down hard, and sued The Magazine, The Publisher, and The General Manager. The judgement that each one signed stated that none of them would ever again aid, abet, or otherwise help Phone Phreaks again, etc. If the Phreaks held their meeting, it might be construed that Byte, and in particular, Byte's Publisher (then General Manager of 73) was aiding Phone Phreaks, and Byte magazine would no longer exist. This was not designated a goodness.

The meeting was cancelled, flyers were printed up to state this, but we were requested not to release details until after the weekend. This request was also honored. However, as we handed the flyers out, we informed the recipient that we intended to cry into our beer over the cancellation at 4PM at the cafeteria.

A happier bunch of melancholy drunks you never saw in your life! About 20-25 people finally showed up, and one thing was unanimous. **THERE WILL BE A PHONE PHREAK CONVENTION WITHIN THE NEXT NINE MONTHS** [REDACTED] Our apologies to those who showed up and then couldn't find us. You, and all our friends will have a chance to get together soon.

Oh yeah, why did we try to have a konklave? Because any Phreak worth his whistle knows that digital is the way of the future. We expected a lot of our friends would be attending anyway and wanted to get together. We'll see you soon.

When I posed for the picture that accompanied the interview article in The Village Voice, reprinted in TAP #48, I had no idea that so many readers and friends would write in to ask where those "Ma Bell Is A Cheap Mother" T-shirts could be bought. The original shirt was made up special for me by a real Bell employee! I find it refreshing to know what Ma Bell employees REALLY think of Ma. As a result of all this reader interest, TAP now has a supply of "Cheap Mother" T's. This T-shirt is available in three colors, yellow, blue, and tan. Sizes - Medium, Large, and Extra Large. Cost: \$4 plus 50¢ postage. Total: \$4.50. Be the first on your block to show Ma what YOU really think of HER!

GET MUGGED !!!

That's right, get mugged!!! Start off the New Year right by getting your TAP mug. The mug is white with black TAP logo, 11 ounces, and stackable. Excellent for coffee, juice, beer or for using as a planter for your Columbian Gold! Cost: \$4 plus 50¢ for postage. Total: \$4.50. If you're a cheap mother yourself, stop by the TAP office on Wednesdays to pick up your mug and save the 50¢ postage charge.

AT&T's Profits Up 17% in Qtr.

American Telephone & Telegraph Co. posted its fifth consecutive billion-dollar quarter yesterday, reporting profits in the three months ended Aug. 31 climbed 17% to \$1.19 billion.

AT&T's revenues in the period rose 10% to \$9.15 billion.

John D. deRutis, chairman, said that while earnings show sustained improvements there are still a number of states "where earnings fall significantly below the level required to support continuing growth and modernization of Bell System facilities."

In these areas, the company will continue to seek rate increases, he said, adding, "in the main, however, our aim is to increase earnings without raising rates."

HELP ROLL BACK PAY PHONE COSTS!

Please do not use this telephone unless your call is of an emergency nature. For more information write: Committee For A Five Cent Phone Call - P.O. Box 30669 - Seattle, Washington 98103.

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Employees Find Ways To Steal

There are 415 known ways to steal money and/or goods from employers and new methods are being discovered almost every day, according to a recent study by the Dale System, Inc.

"A clerk in a busy New York City candy store under-rings every sale by one penny and takes home \$5 a day — for 22 years. A supermarket manager in Oklahoma City sets up his own cash register at peak periods and in three months collects \$75,000 his employers never knew about," noted Dale statisticians.